

We now move into what is arguably the most important technical volume of the entire Tradie platform. This volume defines the complete enterprise data architecture, security model, cloud infrastructure, and DevSecOps strategy required to operate Tradie as a highly available, secure, multi-tenant SaaS platform.

TRADIE MASTER DOCUMENTATION

Volume 09 – Enterprise Data Architecture, Database Design, Security, DevSecOps & Cloud Deployment

Installment 1 – Enterprise Data Architecture & Domain Model

Estimated Final Size: 2,800–3,500 pages

Audience

- Enterprise Architects
 - Database Architects
 - Backend Engineers
 - Cloud Architects
 - Security Engineers
 - DevOps Engineers
 - Platform Engineers
 - Solution Architects
 - Data Engineers
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PART I

Enterprise Data Architecture

Chapter 301

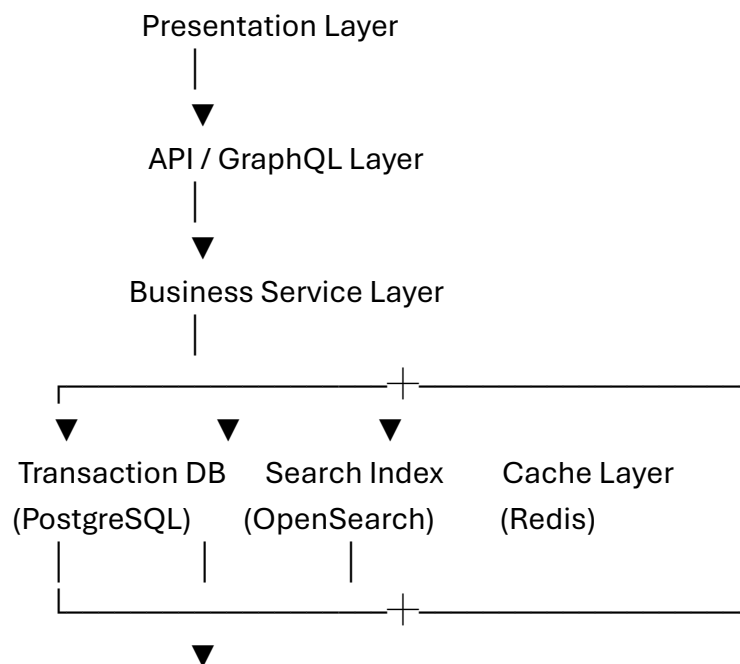
Data Architecture Philosophy

Tradie shall implement a **polyglot persistence architecture**, selecting the most appropriate data technology for each workload while maintaining a unified enterprise data model.

Core Principles

- Domain-Driven Design (DDD)
 - Bounded Contexts
 - Event-Driven Data Synchronization
 - Strong Consistency for Financial Transactions
 - Eventual Consistency for Analytics
 - Multi-Tenant Isolation
 - Horizontal Scalability
 - High Availability
 - Auditability
 - Data Lineage
-

Enterprise Data Layers



Event Streaming
(Kafka)



Data Lake	Warehouse	BI	AI Platform
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Chapter 302

Domain Model

Tradie is organized into the following bounded contexts:

Identity

- Users
- Organizations
- Roles
- Permissions
- Sessions
- Authentication

Marketplace

- Commodity
- Listing
- Auction
- RFQ
- Offer
- Trade

Producer

- Producer

- Farm
 - Plot
 - Crop
 - Harvest
 - Certification
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Warehouse

- Warehouse
 - Bin
 - Inventory
 - Stock Movement
 - Receipt
 - Dispatch
-

Laboratory

- Sample
 - Test
 - Result
 - Certificate
-

Logistics

- Vehicle
- Driver
- Route
- Shipment
- Delivery

Finance

- Invoice
- Payment
- Settlement
- Ledger
- Tax
- Journal

Analytics

- KPI
- Dashboard
- Report
- Forecast
- AI Recommendation

Administration

- Configuration
- Workflow
- Audit
- Notifications
- Feature Flags

Chapter 303

Canonical Entity Relationships

Organization



Chapter 304

Master Data Management (MDM)

Global Masters

- Countries
- States
- Districts
- Villages
- Commodities
- Commodity Varieties
- Grades
- Units of Measure
- Currencies
- Tax Codes
- Incoterms
- Languages

Business Masters

- Warehouses
- Laboratories
- Banks
- Transporters
- Certification Bodies
- Buyers
- Producers
- Commission Agents

Governance Rules

- Version-controlled master records.
- Effective date management.
- Approval workflow for changes.
- Full audit history.
- Soft-delete with archival.

Chapter 305

Database Technology Selection

Workload	Technology
OLTP	PostgreSQL
Caching	Redis
Search	OpenSearch
Event Streaming	Apache Kafka

Workload	Technology
Object Storage	S3-Compatible Storage
Time-Series	TimescaleDB
Analytics Warehouse	ClickHouse
Graph Relationships (Optional)	Neo4j

Chapter 306

Multi-Tenant Data Isolation

Supported tenancy models:

Shared Database, Shared Schema

Suitable for small deployments.

Shared Database, Separate Schemas

Default SaaS deployment.

Dedicated Database per Tenant

For enterprise customers requiring stronger isolation and compliance.

Tenant Resolution

Every request includes:

- Tenant ID
- Organization ID
- Region
- Environment
- Data Residency Policy

Chapter 307

Entity Standards

Every table shall include:

id (UUID or BIGINT)

tenant_id

organization_id

created_at

created_by

updated_at

updated_by

version

status

deleted_at (soft delete)

correlation_id

Chapter 308

Audit Data Model

Every business transaction generates immutable audit records.

Audit Record Structure

Audit ID

Timestamp

User ID

Organization

Module

Action

Entity Type

Entity ID

Before State

After State

IPAddress

Device

Correlation ID

Digital Signature (optional)

Chapter 309

Data Lifecycle Management

States

Draft

↓

Active

↓

Archived

↓

Retention

↓

Deletion

Policies

- Configurable retention periods.
 - Legal hold support.
 - Secure archival.
 - Cryptographic deletion verification where applicable.
-

Chapter 310

Enterprise Metadata Catalog

Each dataset shall define:

- Business owner.
 - Technical owner.
 - Classification.
 - Sensitivity.
 - Lineage.
 - Quality score.
 - Update frequency.
 - Retention policy.
 - Regulatory requirements.
-

PART II

Data Governance

Data Classification

Public

Marketing content

General information

Internal

Operational data

Reports

Configurations

Confidential

Trade information

Financial data

Contracts

Customer information

Restricted

Passwords

Encryption keys

Private certificates

Authentication secrets

Personally identifiable information requiring enhanced protection

Data Quality Rules

Every dataset shall define:

- Completeness
- Accuracy
- Consistency
- Timeliness

- Validity
- Uniqueness

Quality metrics are monitored continuously with automated alerts for threshold breaches.

Acceptance Criteria

- All bounded contexts have clearly defined ownership.
 - Canonical entities are reusable across modules.
 - Multi-tenant isolation is enforced at the data layer.
 - Master data changes are versioned and auditable.
 - Every entity includes standard metadata fields.
 - Data classification and lifecycle policies are applied consistently.
 - Audit records are immutable and traceable.
 - Technology selection aligns with workload characteristics and scalability goals.
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End of Installment 1

This installment establishes the enterprise data architecture, bounded contexts, canonical domain model, master data management strategy, technology selection, multi-tenant data isolation, audit model, lifecycle management, and governance foundation for Tradie.

Preview – Installment 2: Complete Database Schema, ER Diagrams & Physical Data Model (≈1,200–1,500 pages)

The next installment will define:

- Complete logical and physical database schemas.
- Entity-Relationship Diagrams (ERDs) for every bounded context.
- Table definitions with columns, data types, indexes, constraints, and foreign keys.
- Partitioning and sharding strategies.
- Database migration standards.

- Naming conventions.
- Performance optimization and indexing.
- Read/write separation.
- Replication topology.
- Backup, restore, and archival strategies.
- Sample SQL DDL scripts.
- Data validation rules and referential integrity specifications.

This installment will provide the definitive database blueprint for implementing Tradie's persistent storage layer across all modules.

We will now begin the **database implementation blueprint**. This installment moves from conceptual architecture to the physical persistence layer and is intended to be detailed enough for direct implementation by database engineers.

TRADIE MASTER DOCUMENTATION

Volume 09 – Enterprise Data Architecture, Database Design, Security, DevSecOps & Cloud Deployment

Installment 2 – Complete Database Schema, ER Diagrams & Physical Data Model

Estimated Final Size: 1,500–2,000 pages

Audience

- Database Architects
 - PostgreSQL DBAs
 - Backend Engineers
 - Data Engineers
 - Performance Engineers
 - Solution Architects
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PART III

Database Standards

Chapter 311

Naming Conventions

Tables

snake_case

Examples

organizations

users

roles

permissions

commodities

commodity_varieties

farms

warehouse_receipts

inventory_movements

trade_settlements

journal_entries

Columns

snake_case

Examples

organization_id

commodity_id

created_at

updated_at

deleted_at

gross_weight

net_weight

approval_status

Index Names

idx_users_email

idx_inventory_lot

idx_trade_status

idx_invoice_date

Foreign Keys

fk_trade_buyer

fk_trade_seller

fk_inventory_warehouse

fk_receipt_trade

Chapter 312

Common Base Entity

Every transactional table inherits the following structure.

id UUID PRIMARY KEY

tenant_id UUID

organization_id UUID

created_at TIMESTAMP

created_by UUID

updated_at TIMESTAMP

updated_by UUID

version BIGINT

status VARCHAR(30)

deleted_at TIMESTAMP

correlation_id UUID

PART IV

Identity Schema

organizations

id

organization_code

organization_name

legal_name

gst_number

business_type

country

state

district

city

postal_code

address

phone

email

website

status

Indexes

organization_code

gst_number

status

users

id

organization_id

employee_code

first_name

last_name

display_name

email

mobile

password_hash

password_salt

language

timezone

status

last_login

failed_login_attempts

mfa_enabled

Indexes

email

mobile

organization_id

status

roles

id

role_code

role_name

description

system_role

status

permissions

id

permission_code

module

action

resource

description

user_roles

user_id

role_id

assigned_by

assigned_at

Composite Primary Key

(user_id, role_id)

PART V

Commodity Master Schema

commodities

id

commodity_code

commodity_name

commodity_group

hsn_code

scientific_name

default_unit

status

commodity_varieties

id

commodity_id

variety_code

variety_name

origin

quality_standard

status

commodity_grades

id

commodity_id

grade

description

minimum_quality

maximum_quality

PART VI

Producer Schema

producers

id

organization_id

producer_number

producer_name

aadhaar_reference

pan_reference

phone

email

verification_status

farms

id

producer_id

farm_code

survey_number

village

mandal

district

state

country

latitude

longitude

total_area

irrigation_type

plots

id

farm_id

plot_code

area

soil_type

water_source

gps_polygon

harvests

id

plot_id

commodity_id

variety_id

expected_yield

actual_yield

harvest_date

PART VII

Marketplace Schema

listings

id

seller_id

commodity_id

variety_id

grade_id

quantity

available_quantity

unit_price

listing_type

listing_status

expiry_date

rfqs

id

buyer_id

commodity_id

quantity

expected_delivery

budget

status

auctions

id

listing_id

auction_start

auction_end

reserve_price

current_price

status

bids

id

auction_id

bidder_id

bid_amount

bid_time

status

trades

id

buyer_id

seller_id

listing_id

commodity_id

quantity

unit_price

trade_status

contract_reference

PART VIII

Warehouse Schema

warehouses

id

warehouse_code

warehouse_name

warehouse_type

latitude

longitude

capacity

available_capacity

status

warehouse_bins

id

warehouse_id

bin_code

zone

temperature

humidity

capacity

status

warehouse_receipts

id

trade_id

warehouse_id

vehicle_number

gross_weight

tare_weight

net_weight

receipt_number

inventory

id

warehouse_id

bin_id

commodity_id

lot_number

quantity

reserved_quantity

available_quantity

quality_status

inventory_movements

id

movement_type

inventory_id

source_location

destination_location

quantity

movement_time

PART IX

Laboratory Schema

laboratory_samples

id

sample_number

warehouse_receipt_id

collector_id

seal_number

sampling_method

status

laboratory_tests

id

sample_id

parameter

observed_value

unit

acceptable_range

status

quality_certificates

id

certificate_number

sample_id

final_grade

issued_by

issue_date

qr_reference

PART X

Finance Schema

invoices

id

invoice_number

trade_id

invoice_date

subtotal

tax_amount

total_amount

status

payments

id

payment_reference

invoice_id

payment_mode

payment_date

amount

status

journal_entries

id

journal_number

posting_date

reference

status

journal_lines

id

journal_entry_id

account_code

debit

credit

Constraint

$\text{SUM}(\text{debit}) = \text{SUM}(\text{credit})$

settlements

id

trade_id

invoice_id

payment_id

settlement_date

settlement_status

PART XI

Performance Strategy

Partitioning

Large tables to partition by:

- trade_date
 - invoice_date
 - movement_date
 - event_date
 - created_at
-

Read Replicas

Dedicated replicas for:

- Reporting
 - Analytics
 - Dashboard queries
 - AI workloads
-

Index Strategy

- Primary indexes
- Composite indexes

- Partial indexes
 - Covering indexes
 - Full-text indexes (OpenSearch integration)
-

Constraints

- Primary keys
 - Foreign keys
 - Unique constraints
 - Check constraints
 - Exclusion constraints (where applicable)
-

Backup & Recovery

- Daily full backups.
 - Hourly incremental backups.
 - Point-in-time recovery (PITR).
 - Cross-region replication.
 - Periodic restore validation.
-

Acceptance Criteria

- All tables follow the common base entity standard.
- Referential integrity is enforced with foreign keys where appropriate.
- Indexes support expected query patterns.
- Partitioning strategy aligns with data growth.
- Financial tables maintain ACID guarantees.
- Warehouse and inventory schemas preserve traceability.
- Laboratory and quality data maintain end-to-end chain of custody.

- Backup and recovery objectives meet defined RPO/RTO targets.
-

End of Installment 2

This installment defines the foundational physical database schema for Tradie, including identity, commodity, producer, marketplace, warehouse, laboratory, and finance domains, together with naming standards, indexing strategies, partitioning guidance, and operational database practices.

Preview – Installment 3: Security Architecture, DevSecOps, Kubernetes & Cloud Infrastructure (≈2,000–2,500 pages)

The next installment will cover:

- Zero Trust security architecture.
- Identity federation (OAuth2, OIDC, SAML).
- Encryption at rest and in transit.
- Secrets management (e.g., Vault/KMS).
- Key rotation and certificate lifecycle.
- Web Application Firewall (WAF).
- API Gateway security.
- Network segmentation and service mesh.
- Kubernetes cluster architecture.
- Helm charts and deployment topology.
- Infrastructure as Code (Terraform).
- CI/CD pipelines with security gates.
- Container image signing and vulnerability scanning.
- Monitoring, logging, tracing, SIEM integration.
- Multi-region disaster recovery.
- High availability, autoscaling, and production readiness checklists.

This installment will complete the infrastructure blueprint required to deploy Tradie as a secure, resilient, enterprise-grade cloud platform.

We now begin the infrastructure and operational security blueprint that enables Tradie to be deployed as a secure, resilient, enterprise-grade SaaS platform.

TRADIE MASTER DOCUMENTATION

Volume 09 – Enterprise Data Architecture, Database Design, Security, DevSecOps & Cloud Deployment

Installment 3 – Security Architecture, DevSecOps, Kubernetes & Cloud Infrastructure

Estimated Final Size: 2,200–2,800 pages

Audience

- Cloud Architects
 - DevSecOps Engineers
 - Security Architects
 - Platform Engineers
 - Infrastructure Engineers
 - SRE Teams
 - Compliance Officers
-

PART XII

Enterprise Security Architecture

Chapter 313

Security Philosophy

Tradie follows a **Zero Trust Architecture** where:

- No user, device, service, or network is inherently trusted.
- Every request is authenticated, authorized, encrypted, and continuously evaluated.

- Security is embedded throughout the software development lifecycle (Shift Left Security).

Core Principles

- Verify Explicitly
 - Least Privilege
 - Assume Breach
 - Defense in Depth
 - Continuous Monitoring
 - Secure by Default
-

Chapter 314

Identity & Access Management (IAM)

Authentication

Supported mechanisms:

- OAuth 2.1
- OpenID Connect (OIDC)
- SAML 2.0
- Multi-Factor Authentication (MFA)
- Passwordless Authentication (FIDO2/WebAuthn)
- Biometric Authentication (Mobile)

Identity Providers

- Microsoft Entra ID
- Google Workspace
- Okta
- Keycloak
- Auth0

- LDAP / Active Directory
-

Authorization

Layers:

- Role-Based Access Control (RBAC)
 - Attribute-Based Access Control (ABAC)
 - Policy-Based Access Control (PBAC)
 - Row-Level Security (RLS) for multi-tenant data isolation
-

Chapter 315

Cryptography Standards

Encryption in Transit

- TLS 1.3 minimum
- Perfect Forward Secrecy (PFS)
- Strong cipher suites
- Mutual TLS for service-to-service communication where required

Encryption at Rest

- AES-256 for databases
 - AES-256 for object storage
 - Encrypted backups
 - Encrypted snapshots
-

Key Management

Supported services:

- AWS KMS
- Azure Key Vault

- Google Cloud KMS
- HashiCorp Vault
- Hardware Security Modules (HSM) where required

Key Rotation

- Automatic rotation for application keys
 - Certificate renewal automation
 - Emergency key revocation procedures
-

PART XIII

Application Security

Chapter 316

Secure Coding Standards

Adopt:

- OWASP ASVS
- OWASP Top 10
- CWE/SANS Top 25

Mandatory protections include:

- Input validation
- Output encoding
- Parameterized SQL queries
- CSRF protection
- XSS prevention
- SSRF prevention
- Command injection prevention
- Secure file upload validation

Secrets Management

Secrets must never be stored in source code.

Supported secret types:

- Database credentials
- API keys
- OAuth client secrets
- Certificates
- Encryption keys
- Third-party integration credentials

Chapter 317

API Security

All APIs require:

- JWT validation
- Token expiry checks
- Audience validation
- Scope validation
- Rate limiting
- Replay protection
- Idempotency (where applicable)

API Gateway Responsibilities

- Authentication
- Authorization
- Request validation
- Schema enforcement

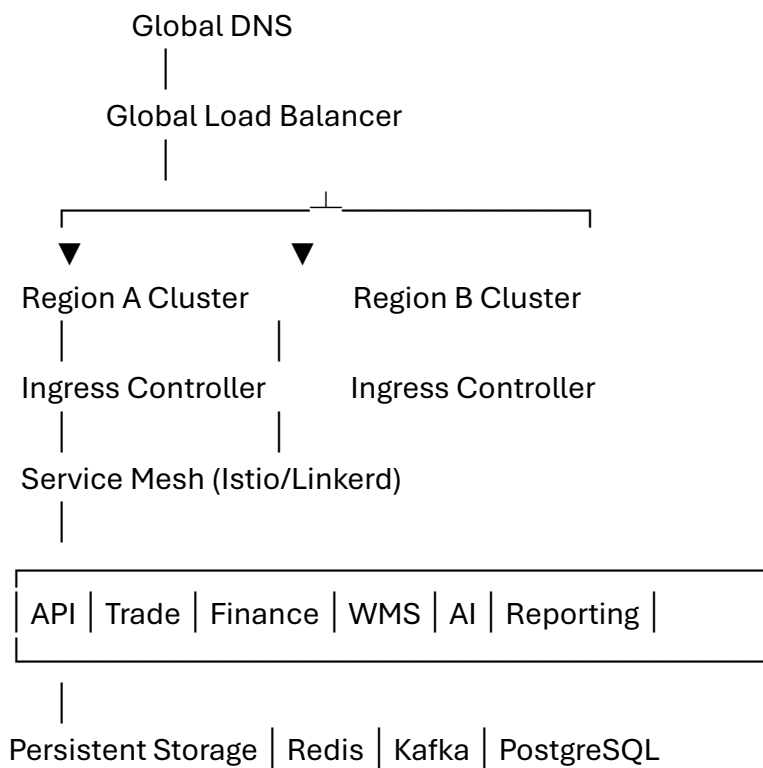
- Traffic shaping
- WAF integration
- DDoS mitigation
- Request logging

PART XIV

Kubernetes Platform

Chapter 318

Cluster Architecture



Namespace Strategy

- platform
- identity

- marketplace
 - warehouse
 - logistics
 - finance
 - ai
 - monitoring
 - security
 - integrations
-

Workload Types

- Deployments
 - StatefulSets
 - DaemonSets
 - Jobs
 - CronJobs
-

Chapter 319

Service Mesh

Capabilities:

- mTLS
- Traffic routing
- Canary releases
- Blue/Green deployments
- Circuit breaking
- Retry policies
- Distributed tracing

- Service authorization

PART XV

DevSecOps Pipeline

Chapter 320

CI/CD Workflow

Developer Commit

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Static Code Analysis

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Dependency Scan

↓

Unit Tests

↓

Build

↓

Container Image Scan

↓

Integration Tests

↓

Security Tests

↓

Performance Tests

↓

Approval Gate

↓

Deploy to Staging

↓

User Acceptance Testing

↓

Production Deployment

Quality Gates

Build proceeds only if:

- Code coverage threshold met
 - No critical vulnerabilities
 - No license violations
 - Performance benchmarks passed
 - Infrastructure validation passed
-

Chapter 321

Infrastructure as Code (IaC)

Supported tools:

- Terraform
- Helm
- Kustomize
- Ansible (where appropriate)

Principles

- Immutable infrastructure
 - Version-controlled environments
 - Automated provisioning
 - Policy-as-Code
 - Environment parity
-

PART XVI

Monitoring & Observability

Chapter 322

Monitoring Stack

Recommended components:

- Prometheus
 - Grafana
 - OpenTelemetry
 - Jaeger
 - Loki
 - Alertmanager
-

Metrics Categories

- API latency
 - Error rates
 - Throughput
 - Database performance
 - Cache hit ratio
 - Kafka lag
 - Kubernetes health
 - Business KPIs
-

Logging

Centralized structured logging with:

- Correlation IDs
 - Trace IDs
 - User context
 - Severity levels
 - Retention policies
-

Chapter 323

Security Monitoring

Integrations with SIEM platforms such as:

- Microsoft Sentinel
- Splunk
- IBM QRadar
- Elastic Security

Detectable Events

- Failed logins
- Privilege escalation
- Suspicious API usage
- Configuration changes
- Data export activity
- Malware indicators

PART XVII

Disaster Recovery & Business Continuity

Chapter 324

Backup Strategy

- Continuous WAL archiving for PostgreSQL
 - Daily full backups
 - Hourly incremental backups
 - Immutable backup storage
 - Cross-region replication
-

Disaster Recovery Targets

Metric	Target
Recovery Point Objective (RPO)	≤ 15 minutes
Recovery Time Objective (RTO)	≤ 2 hours

High Availability

- Multi-AZ database clusters
 - Multi-region failover
 - Redundant API gateways
 - Highly available Kafka clusters
 - Distributed object storage
 - Automatic health checks and failover
-

Chapter 325

Production Readiness Checklist

Before production deployment:

- Security assessment completed
- Penetration testing passed
- Disaster recovery test successful
- Backup restoration verified
- Performance benchmarks achieved
- Monitoring dashboards configured
- Alerting rules validated
- Runbooks documented
- On-call procedures established

- Compliance review completed
-

Acceptance Criteria

- Zero Trust controls implemented across all layers.
 - Secrets managed externally with automated rotation.
 - Kubernetes workloads deployed with least-privilege permissions.
 - CI/CD pipelines enforce security and quality gates.
 - Infrastructure reproducible through Infrastructure as Code.
 - Monitoring, tracing, and logging provide end-to-end observability.
 - Disaster recovery objectives meet defined RPO/RTO targets.
 - Production environments validated through readiness assessments and recovery testing.
-

End of Installment 3

This installment defines the security architecture, identity and access management, cryptographic standards, application security controls, Kubernetes platform design, DevSecOps pipeline, Infrastructure as Code strategy, observability stack, and disaster recovery architecture required for Tradie.

Recommended Next Major Volume

At this point, the documentation covers business vision, UX, APIs, data architecture, and infrastructure. The next logical volume would be:

Volume 10 – Enterprise Implementation, Testing, Operations & Product Delivery, including:

- Backend implementation standards.
- Frontend architecture (Flutter, React, React Native).
- Coding standards and repository structure.
- Automated testing strategy (unit, integration, E2E, performance, security).
- QA automation framework.

- Release management.
- Feature flag strategy.
- Operational runbooks and SOPs.
- Incident management.
- User training manuals.
- Administrator guides.
- Deployment and migration playbooks.
- Production support procedures.
- Product roadmap and future extensibility.

This would complete the documentation set into a comprehensive enterprise blueprint suitable for design, development, deployment, operations, and long-term maintenance.